A LINEAR COMBINATION OF ELEMENTS DEFINING AN INJECTIVE MAP IS A NONZERODIVISOR IN THE POLYNOMIAL RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Associated primes of a module over a Noetherian ring and the criterion that a map of finite modules is injective if it is injective after localising at every associated prime. the author cites Algebra, Lemma algebra-lemma-zero-at-ass-zero at proof lines 7787-7789. Supplies the localisation criterion that turns the verification at associated primes into injectivity.
- Symmetric powers of a free module are free, and the multiplication by a linear form on them. used at proof lines 7772-7786. Provides the maps whose injectivity is checked after localisation.
- Every ring is a filtered union of Noetherian subrings, and injectivity is preserved by such unions. used at proof lines 7791-7795. Reduces the general case to the Noetherian one.

Source reference. Original lemma 068P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8013–8013. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0311: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3968–3968. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-McCoy: sources/stacks-more-algebra/pinned-original-source.tex, source lines 7757–7757. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let R be a ring. Let $a_1, \ldots, a_n \in R$ be elements such that $R \to R^{\oplus n}$, $x \mapsto (xa_1, \ldots, xa_n)$ is injective. Then the element $\sum a_i t_i$ of the polynomial ring $R[t_1, \ldots, t_n]$ is a nonzerodivisor.*

Proof. If one of the a_i is a unit this is just the statement that any element of the form $t_1 + a_2 t_2 + \ldots + a_n t_n$ is a nonzerodivisor in the polynomial ring over R.

Case I: R is Noetherian. Let $\mathfrak{q}_j$, $j = 1, \ldots, m$ be the associated primes of R. We have to show that each of the maps

$$\sum a_i t_i : \mathrm{Sym}^d(R^{\oplus n}) \longrightarrow \mathrm{Sym}^{d+1}(R^{\oplus n})$$

is injective. As $\mathrm{Sym}^d(R^{\oplus n})$ is a free R-module its associated primes are $\mathfrak{q}_j$, $j = 1, \ldots, m$. For each j there exists an $i = i(j)$ such that $a_i \notin \mathfrak{q}_j$ because there exists an $x \in R$ with $\mathfrak{q}_j x = 0$ but $a_i x \neq 0$ for some i by assumption. Hence a_i is a unit in $R_{\mathfrak{q}_j}$ and the map is injective after localizing at $\mathfrak{q}_j$. Thus the map is injective, see Algebra, Lemma 0311.

Case II: R general. We can write R as the union of Noetherian rings R_λ with $a_1, \ldots, a_n \in R_\lambda$. For each R_λ the result holds, hence the result holds for R. $\square$